

<https://doi.org/10.1038/s43247-025-02544-2>

Microbial communities and litter quality co-mediate seagrass litter decomposition under eutrophication



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The decomposition of recalcitrant organic matter (ROM) in seagrass litter plays a crucial role in mediating blue carbon magnitude. However, the mechanisms underlying ROM decomposition in seagrass litter under eutrophication, driven by anthropogenic pressures, remain poorly understood. Here, we investigated the effects of eutrophication on the decomposition of two types of seagrass litter (leaf and sheath) through laboratory experiments adding nitrogen and macroalgae. We found that macroalgae addition inhibited ROM breakdown in leaves and sheaths. Nitrogen addition enhanced ROM decomposition in leaves by 4% while suppressing it in sheaths compared to the treatments without nitrogen addition. Additionally, the combined addition of nitrogen and macroalgae led to a more substantial loss of sheath ROM (35–39%) compared with macroalgae alone. The opposite response of ROM breakdown in leaf and sheath under nitrogen addition were co-regulated by litter quality and bacteria taxa (i.e., Flavobacteriales, Actinomarinales, and Phycisphaerales). In the presence of macroalgae, fungi, including *Lecythophora luteoviridis*, *Tyrannosorus hystrioides*, and *Pseudeutotium ovale*, were found to play a significant role in the decomposition of leaf ROM, but not in sheath decomposition. These findings highlight the negative effects of eutrophication on litter ROM sequestration and the important role that litter quality and microbe play.

Seagrass meadows rank among the most productive ecosystems, with an average annual production of $1.012 \text{ kg DW m}^{-2} \text{ year}^{-1}$ ¹. Although only occupying less than 0.2% of the area of global ocean, seagrass ecosystems contribute roughly 10% of marine organic carbon sequestration². Over 50% of the organic carbon storage in these meadows originates from seagrass, with above-ground vegetative tissues (leaf and sheath/stem) contributing about half of the total living biomass, making them significant contributors to carbon storage^{1,3}. The abscission of leaf and sheath occurs after senescence, with a substantial portion either exported to adjacent ecosystems or buried within the sediment *in situ*^{4,5}, accounting for over 40% of seagrass primary production⁶. Leaf and sheath litter release organic carbon through early leaching and microbial degradation, influencing carbon (C) and nitrogen (N) cycling within ecosystems^{7,8}. Therefore, seagrass leaf and sheath decomposition and their role in carbon sequestration are increasingly gaining attention from researchers^{9–11}.

Anthropogenic N inputs from agriculture, aquaculture, and coastal development have been extensively documented to lead to nutrient enrichment and macroalgae proliferation in coastal areas^{12,13}. This phenomenon is referred to as coastal eutrophication. Globally, seagrass ecosystems have declined at a rate of 7% y⁻¹ since 1990, with 29% of their pre-2009 distribution already lost, primarily due to eutrophication¹⁴. Beyond directly triggering seagrass loss, nutrient enrichment can impact the decomposition dynamic of above-ground seagrass tissues, thereby altering carbon cycling processes and blue carbon storage potential^{15–18}. For example, Apostolaki et al.¹⁹ observed a 30% enhancement in the decomposition rate of *Posidonia oceanica* leaves when subjected to high seawater nutrient conditions. However, Qin et al.²⁰ demonstrated that nutrient enrichment reduced the decomposition rate of *Zostera marina* leaf by ~5%. Further, the impact of nutrient addition on litter decomposition is associated with various factors, including litter quality (N, P, C/N, amino acid, lignin, et al.),

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ambient nutrient levels, fertilization rate, and may vary across different decomposition stages^{21,22}. In addition to direct impacts of inorganic nutrient addition, macroalgae litter production after a macroalgae bloom-collapse can impose substantial labile organic carbon on seagrass meadows and may indirectly influence seagrass litter decomposition rate⁷. Macroalgal blooms are increasing in coastal regions worldwide, including nutrient over-enrichment of shallow estuaries, bays, and lagoons^{23,24}. Thus, in the field, the N load with co-occurring macroalgal blooms represents a natural scenario. However, while existing studies have primarily focused on the influence of nitrogen load and macroalgal bloom on litter decomposition, respectively, there is an urgent need to investigate the interactive effects of N and macroalgae on seagrass litter decomposition.

The external fresh organic carbon (OC) (e.g., including easily available organic substances) addition exacerbated native OC loss via the stimulation of microbial remineralization, i.e., the 'microbial priming effect (PE)'²⁵. The PE is a phenomenon in which inorganic N/P or OC input stimulates microbes to metabolize native OC²⁶. This phenomenon is also observed in seagrass ecosystems²⁷. The input of fresh carbon can promote (i.e., positive priming) or inhibit (i.e., negative priming) the decomposition of recalcitrant organic matter (ROM) in soil and seagrass litter, ultimately affecting the stability and turnover of ecosystem C sinks^{7,28}. Recalcitrant organic matter, mainly refers to cellulose- and lignin-associated organic matter, is part of total organic matter that is highly resistant to decomposition by microorganisms and chemical processes in the environment²⁹. The co-metabolic mechanism behind the positive PE is that microorganisms synthesize ROM decomposition enzymes under OC input to promote litter decay⁷. However, there is no direct evidence of which microorganisms play an important role, either in positive or negative PE. Further, both the exogenous organic substrates types (i.e., labile and recalcitrant) and soil properties (e.g., N, P content) are also considered to be important factors affecting the intensity and direction of the PE within terrestrial ecosystems^{26,30,31}. The PE of different organic substrates inputs (i.e., microalgae and seagrass) on sediment OC have been studied, but the effects of substance (i.e., litter quality) on PE are not well understood in seagrass meadows. Compared with other tissues, seagrass leaves are rich in protein and lipids²⁹. The sheath of seagrass is mainly composed of p-HBA phenolic substances, carbohydrates, and cellulose³². Macroalgae, which are mainly composed of polysaccharides and lack lignocellulose³³. It is necessary to study whether there is a different PE of seagrass leaves and sheath in response to macroalgae and N input.

In this study, our aim is to investigate the microbial taxa and litter quality of the effects of macroalgae and N loading on PE of litter decomposition. Elucidating the impacts of eutrophication on seagrass litter decomposition and carbon cycling is critical for quantifying anthropogenic threats to seagrass ecosystems. Specifically, our approach is to perform a laboratory incubation using *Thalassia hemprichii*, a tropical seagrass species that faces threats from ammonium toxicity and macroalgal blooms^{18,34}. We applied both inorganic N and fresh *Ulva pertusa* to the leaf and sheath of *T. hemprichii* to quantify a PE over 116 days of decomposition. We assessed the changes of labile and recalcitrant OM types, as well as the microbial community structure of the seagrass litter throughout the incubation period. We formulated the following hypotheses: the direction of microbial PE of seagrass leaves and sheaths exhibit different patterns in the addition of N and macroalgae.

Results

Changes of litter decomposition

Leaf litter mass decreased continuously, and the remaining mass was 29.5% to 42.6% at the end of the experiment (Fig. 1a). In contrast, sheath mass loss plateaued after 65 days, with 82.1–92.0% mass remaining after the decomposition (Fig. 1b). Although N addition exhibited opposing effects on leaf (−11%) and sheath (+4%) mass remaining, no statistically significant differences were detected (leaf: $F = 1.952$, $p = 0.185$; sheath: $F = 0.119$, $p = 0.889$) (Supplementary Table 1). Macroalgae addition both decreased leaf and sheath decay (leaf: $F = 7.106$, $p = 0.021$; sheath: $F = 5.325$, $p = 0.040$)

(Supplementary Table 1 and Fig. 1b). The addition of macroalgae did not significantly affect the remaining mass of leaf litter during the initial 37 days (15 d: $F = 0.519$, $p = 0.485$; 37 d: $F = 0.177$, $p = 0.651$) but notably inhibited the decomposition process thereafter (65 d: $F = 12.919$, $p = 0.004$; 116 d: $F = 5.406$, $p = 0.038$) (Supplementary Tables 2 and 3). Macroalgae addition exerted inhibitory effect on sheath decay at 37 days (Supplementary Tables 4 and 5). The interactive effect of N and macroalgae was significant on sheath mass remaining at 37 days and 116 days (37 d: $F = 5.116$, $p = 0.025$; 116 d: $F = 4.592$, $p = 0.033$), while no significant interactive effects were observed on leaf mass remaining (Supplementary Tables 1 and 4). After 65 days of decomposition, the remaining mass of macroalgae in leaf and sheath groups approached 0% (Supplementary Fig. 1). The decomposition coefficients (k) of the leaf and sheath in macroalgae addition treatments exhibited significant reductions of 19% and 41% (mean values), respectively, compared to the treatments without macroalgae addition (Table 1).

Changes of biochemical compositions of litter

For the leaf, C and N exhibited a dynamic pattern of leaching, enrichment, and release, except for the N in treatments without macroalgae addition (Supplementary Fig. 2a, b). N in the sheath showed a net release of N before 15 days, followed by an enrichment phase, and ultimately stabilized or remained enriched by the end of the experiment, except in the macroalgae addition treatments (Supplementary Fig. 2e). Concerning the C/N ratio, the interactive effect of N and macroalgae significantly influenced the C/N of the leaf, whereas showed no significant effect on the sheath (Supplementary Tables 1 and Supplementary Fig. 2c, g). The interactive effect of macroalgae and N had no significant effect on the remaining proportion of leaf total phenol, but macroalgae addition inhibited the loss of sheath total phenol (Supplementary Fig. 2d, h and Supplementary Tables 1, 2, 4, 5).

Leaves, sheaths, and macroalgae manifested similar Z-shaped TGA curves (Supplementary Fig. 3a). Analysis of the rate of change indicated that the primary differences in mass loss occurred between 200 °C and 400 °C (Supplementary Fig. 3b). The initial TI_1/OM and TI_2/OM in the leaf were higher than those in the sheath, whereas TI_3/OM showed the opposite pattern (higher in the sheath) (Supplementary Table 6). As the leaf and sheath decomposed, the remaining proportions of TI_1 , TI_2 , and TI_3 gradually decreased (Supplementary Fig. 4). In the leaf, the addition of N promoted the loss of OM, especially TI_2 ($F = 4.455$; $p = 0.036$) (Supplementary Fig. 4a–c and Supplementary Tables 1, 2, 7). Additionally, the decomposition rate of TI_1 , TI_2 , and TI_3 in leaf of N treatments was higher than that of control (Table 1). N addition, increased the remaining proportions of TI_1 , TI_2 , and TI_3 in the sheath, although no significant differences were observed (Supplementary Fig. 4d–f and Supplementary Table 1). The addition of macroalgae significantly inhibited the decomposition of TI_1 , TI_2 , and TI_3 in the leaf and sheath (Table 1 and Supplementary Tables 1–5). The remaining proportions and decomposition rates of TI_1 , TI_2 , and TI_3 in the leaf and sheath in N combined with macroalgae addition treatments fell between the macroalgae addition alone and the N addition alone treatments (Table 1 and Supplementary Fig. 4). The interactive effect of N and macroalgae was significant of TI_3 remaining at 37 and 116 days in the sheath (Supplementary Tables 4 and 8).

Changes of microbial communities and enzyme activity

Bacterial ASV richness for leaves increased over time, peaking at 65 days and declining thereafter, except for the control, macroalgae combined with low N or high N addition treatments (Supplementary Fig. 5a). Sheath-associated bacterial ASV richness peaked at 37 days except for the high N and macroalgae combined with high N treatments (Supplementary Fig. 5b). The interactive effect of nitrogen and macroalgae on bacterial ASV richness in leaves was significant at 37 and 116 days (Supplementary Table 9). Nitrogen addition significantly increased bacterial ASV richness (measured at 65 days) in sheaths (Supplementary Table 10). Rhizobiales (leaf: 17.68% vs.

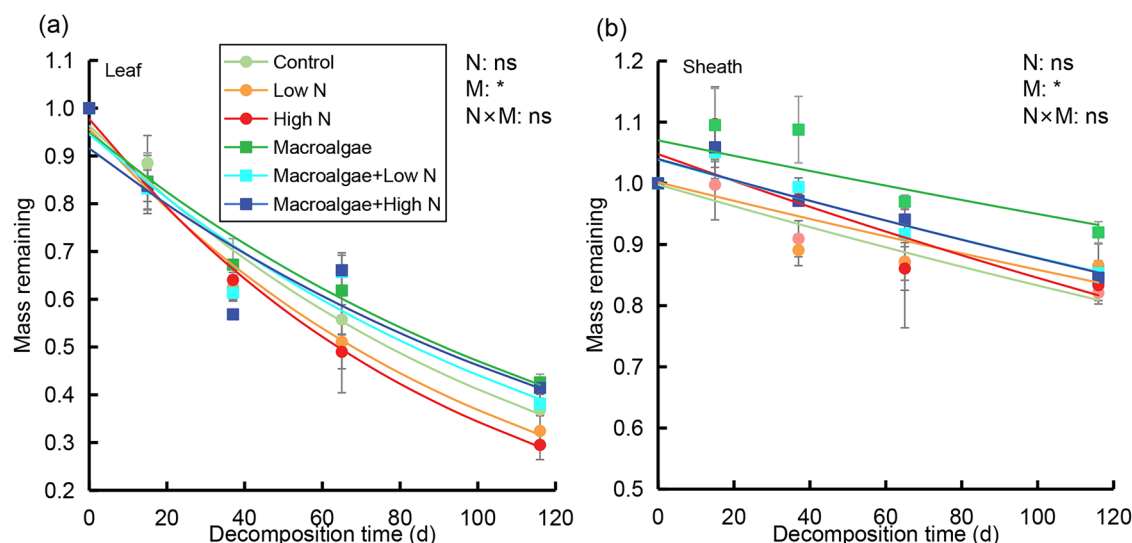


Fig. 1 | Dynamics of leaf and sheath mass remaining of *T. hemprichii* under different nitrogen and macroalgae addition treatments. a Leaf mass remaining; b sheath mass remaining. Data are shown as mean $\pm$ standard error ($n = 3$). N nitrogen addition, M macroalgae addition, ns not significant ($p > 0.05$); *, $0.01 < p \leq 0.05$.

Table 1 | The decomposition coefficients (k , k_{T11} , k_{T12} , k_{T13} , mean) of litter mass, T_{11} , T_{12} , and T_{13}

Tissues	Treatment	Decomposition constant k (day^{-1})	R^2	$T_{50\%}$ (d)	$T_{95\%}$ (d)	k_{T11}	k_{T12}	k_{T13}
Leaf	Control	0.0094	0.9695	73	319	0.0233	0.0197	0.0105
	Low N	0.0107	0.9852	65	280	0.0239	0.0208	0.0114
	High N	0.0111	0.9990	62	270	0.0253	0.0214	0.0117
	Macroalgae	0.0080	0.9575	87	374	0.0196	0.0153	0.0082
	Macroalgae + Low N	0.0086	0.8965	81	348	0.0209	0.0153	0.0088
	Macroalgae + High N	0.0085	0.8231	82	352	0.0228	0.0172	0.0097
Sheath	Control	0.0019	0.9336	365	1577	0.0045	0.0065	0.0030
	Low N	0.0015	0.6413	462	1997	0.0044	0.0059	0.0024
	High N	0.0016	0.6297	433	1872	0.0047	0.0062	0.0029
	Macroalgae	0.0005	0.3098	1386	5991	0.0028	0.0034	0.0012
	Macroalgae + Low N	0.0012	0.7328	578	2496	0.0034	0.0048	0.0021
	Macroalgae + High N	0.0012	0.7208	578	2496	0.0037	0.0046	0.0020

The model-estimated times to 50% ($T_{50\%}$) and 95% ($T_{95\%}$) decomposition of mass were also presented.

Sheath: 18.17%), Flavobacteriales (leaf: 15.69% vs. Sheath: 12.30%), and Rhodobacteriales (leaf: 14.59% vs. Sheath: 11.97%) were the dominated bacterial orders across all samples during leaf and sheath decomposition (Supplementary Fig. 6). Fungal ASV richness of seagrass leaves increased notably during the first 15 days of decay and then decreased in macroalgae addition and macroalgae combined with N treatments, while in sheath, it decreased continuously (Supplementary Fig. 5c, d). Throughout the decomposition of leaves and sheath, Saccharomycetales, Hypocreales, and Eurotiales were consistently present across all treatments and time points (Supplementary Fig. 7).

In the plot of NMDS ordination, microbial communities were distinctly separated by different treatments except fungal communities associated with the leaf and sheath (Fig. 2). The differences in leaf and sheath bacterial community composition under N addition were significant (leaf: $R^2 = 0.039$, $p = 0.001$; sheath: $R^2 = 0.043$, $p = 0.001$) (Fig. 2a, b and Supplementary Table 11). The SIMPER analysis revealed that the top 10 species contributing most to the differences in the community composition explained 7.44%–8.65% of the differences. Notably, in leaves, a *Euzeyella* ASV and a *Actinomarinales* ASV were more abundant in the High N addition treatments compared to the control (Supplementary Table 12). In sheath, a *Urania_1B_19_marine_sediment_group* ASV exhibited greater

abundance in the control than in the N addition treatments (Supplementary Table 12).

Significant differences in bacterial community composition within leaves and sheaths were also observed between macroalgae addition treatments (leaf: $R^2 = 0.119$, $p = 0.001$; sheath: $R^2 = 0.102$, $p = 0.001$) (Fig. 2a, b and Supplementary Table 11). A *Sva0996_marine_group* ASV and a *Woeseia* ASV were more abundant in the macroalgae addition treatments compared to the treatments without macroalgae addition for both leaf and sheath (Supplementary Table 13). For leaf, the fungal community composition also varied significantly with the addition of macroalgae ($R^2 = 0.112$, $p = 0.001$) (Fig. 2c and Supplementary Table 14). *Galactomyces_pseudocandidus*, a *Fusarium* ASV, *Lecythophora_luteoviridis*, *Psathyrella_lutensis*, *Tyranosorus_hystrioides*, and *Pseudeurotium_ovale* were more abundant in treatments without macroalgae than in macroalgae addition treatments (Supplementary Table 13).

Significant differences in leaf and sheath bacterial community composition were also observed between N combined with macroalgae addition treatments (leaf: $R^2 = 0.037$, $p = 0.001$; sheath: $R^2 = 0.045$, $p = 0.001$) (Fig. 2a, b and Supplementary Table 11). For leaf, the relative abundance a *Sva0996_marine_group* ASV was lower in combined macroalgae and N addition than that in macroalgae alone. For sheath, the

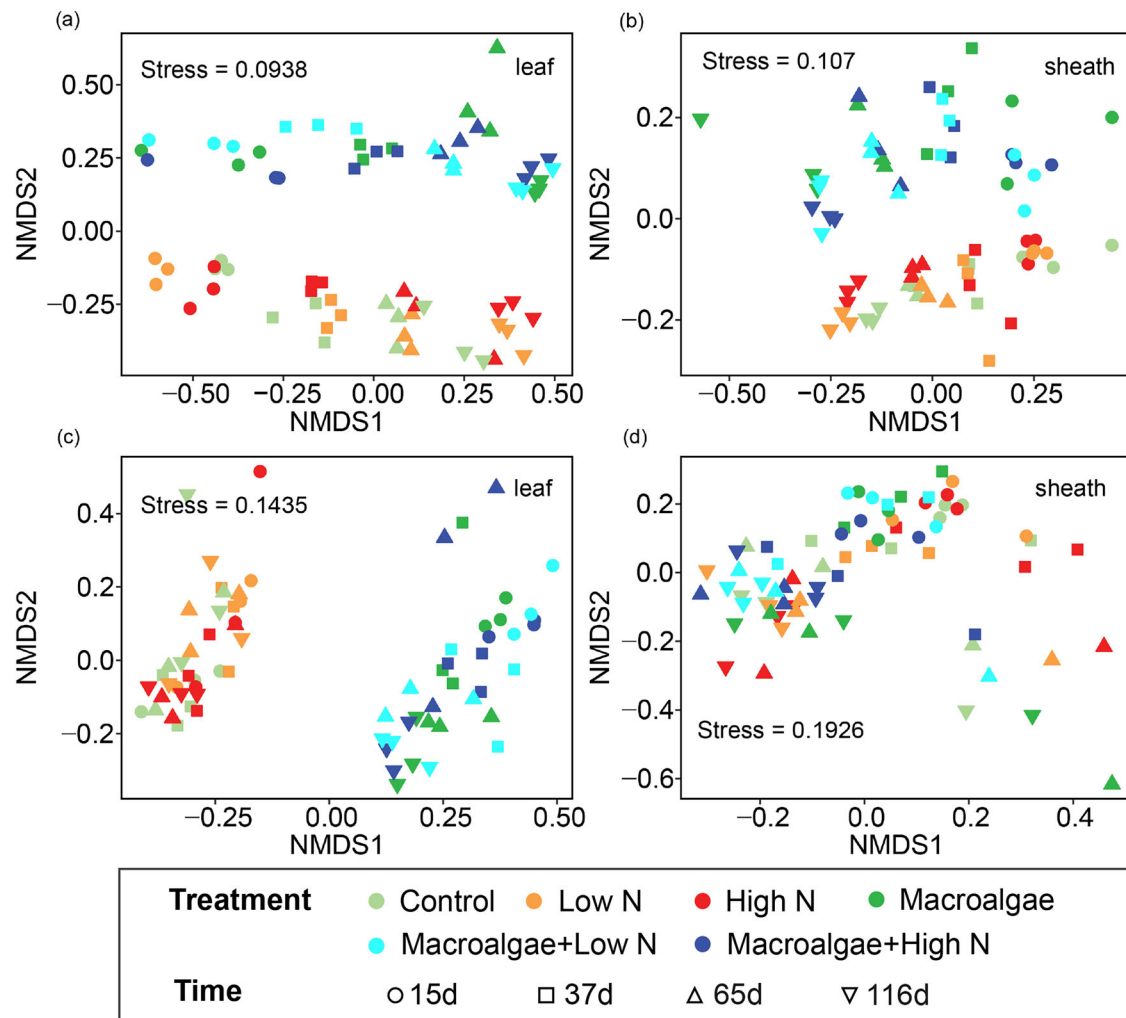


Fig. 2 | The two-dimensional nonmetric multidimensional scaling (NMDS) ordinations show dissimilarity of litter microbial communities between N and macroalgae addition treatments at different decay time. **a** leaf bacteria; **b** sheath bacteria; **c** leaf fungi; **d** sheath fungi.

abundance of a *Urania_1B_19_marine_sediment_group* ASV was higher in Control than that in Macroalgae treatment. A *Cellulophaga* ASV, an *Actinomarinales* ASV, and a *Candidatus_Berkiella* ASV were more abundant in N combined with macroalgae addition treatments compared with Low N and High N treatments. Moreover, a *Actinomarinales* ASV, a *Woeseia* ASV, *Tamlana_crocina*, and a *Candidatus_Berkiella* ASV were more abundant in macroalgae combined with High N compared with Macroalgae treatment (Supplementary Table 15).

The nitrogen, macroalgae, and their interaction did not exhibit significant effects on β -glucosidase (β G) activity in both leaves and sheaths at 37 days of decomposition ($p > 0.05$) (Supplementary Fig. 8 and Supplementary Tables 2 and 4). Compared with the Control, Low N and High N additions increased β G activity in the leaf by 55–58%. Both nitrogen and macroalgae additions decreased β G activity in the sheath by 28–50% and 44%, respectively.

Correlations between microbial composition and litter biochemical composition during decomposition

Our RDA ordination of bacterial species revealed significant correlations between litter biochemical components (C, N, TI₁, TI₂, TI₃, and phenol) and bacterial community composition ($p < 0.05$) (Fig. 3a, b). In the case of leaf, the relative abundances of a Gammaproteobacteria ASV, a Dadabacteriales ASV, and a Rhizobiales ASV were positively correlated with the TI₃ and C. Conversely, the relative abundances of these species were negatively correlated with TI₁ and phenol (Fig. 3a). The TI₁, TI₂, TI₃, C, and N of sheath

were positively correlated with the relative abundance of a Chlamydiales ASV, a Dadabacteriales ASV, a Rhizobiales ASV, a *Sva0996_marine_group* ASV, and a *Actinomarinales* ASV, while negatively with a Rhodobacteraceae ASV and a *Arenibacter* ASV (Fig. 3b). The RDA of the relationship between the fungal community structures of the leaf and sheath and litter biochemical components was poorly interpreted (<10% explained variation) (Fig. 3c, d).

Pathways of nitrogen and macroalgae addition affecting ROM decomposition

The results of the Partial Least Squares Path Model (PLS-PM) analysis (GOF = 0.4829 and 0.5226) explained 68.4% of the total variation in litter ROM remaining of leaf with goodness of fit (GOF) 0.4829, while 42.4% with GOF 0.5226 of sheath. Carbon quality and microbial communities were the most important factors that directly explained changes in the leaf ROM remaining (path coefficient = 0.5356 and 0.4501) (Fig. 4a, c). Further, PLS-PM analysis confirmed the negative effect of N addition on leaf ROM remaining (including direct and total effects), while revealing a positive effect on sheath ROM remaining (Fig. 4b, d). Although macroalgae addition had a negative direct effect on leaf ROM remaining (direct path coefficient = −0.33), the total effect was positive, indicating the indirect inhibition effect on ROM remaining through carbon quality and microbial community (Fig. 4a, b). A large positive effect on the sheath ROM was caused by the application of macroalgae by directly (direct path coefficient = 0.72) (Fig. 4c, d). It is noteworthy that both macroalgae and N addition negatively

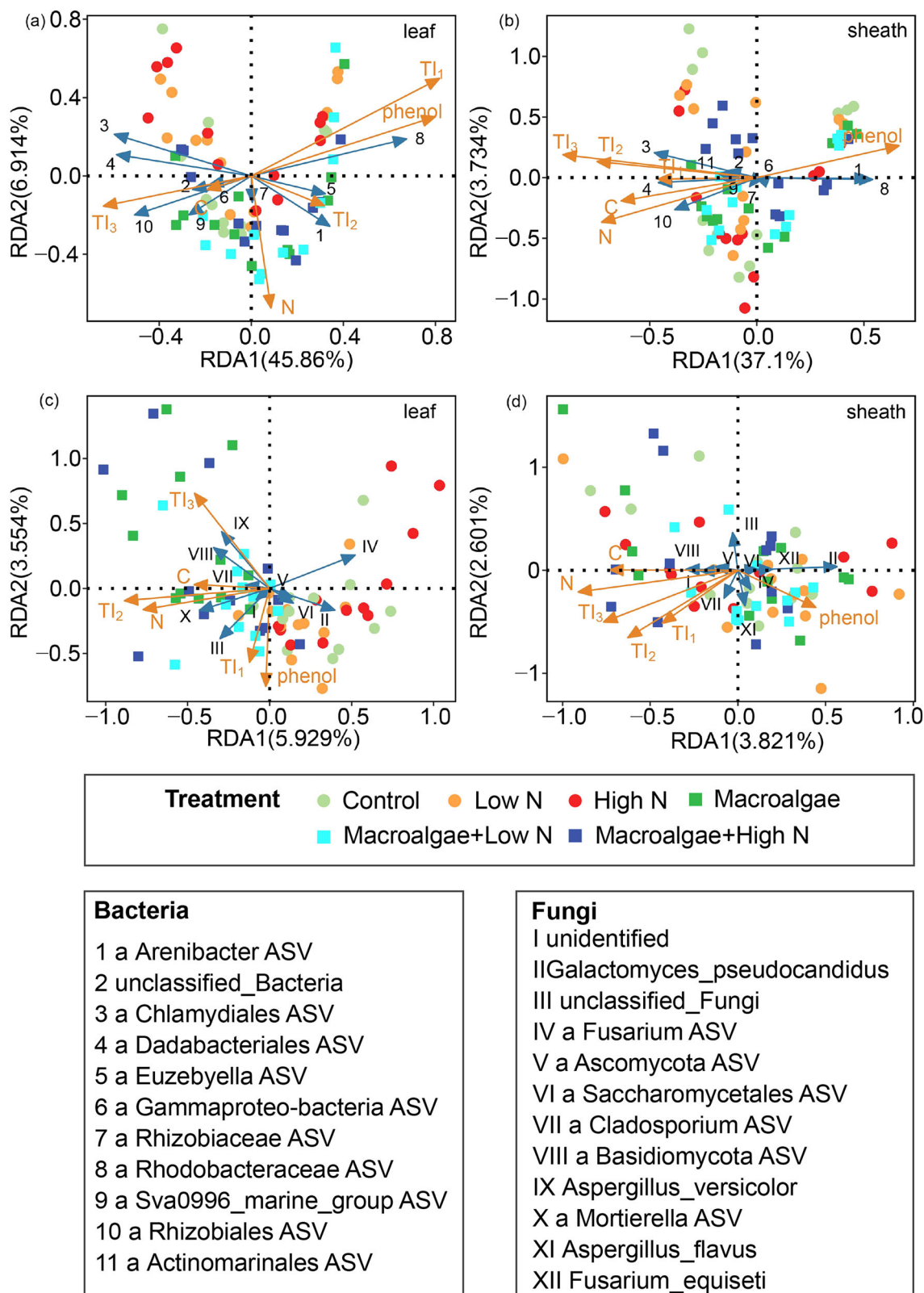


Fig. 3 | Redundancy analysis to determine correlations between the relative abundance of microbial species (Top 10) and temporal dynamics of litter biochemical composition (C, N, phenol, TI₁, TI₂, TI₃). Correlations between bacterial

species and leaf and sheath biochemical composition (a, b) fungal species and leaf and sheath biochemical composition (c, d). Arabic numerals denote different bacterial species, while Roman numerals denote different fungal species.

impacted sheath ROM remaining, with a negligible effect on leaf ROM remaining.

We applied random forest regression modeling and SHAP analysis to identify the major microbial taxa regulating ROM decomposition. The most

important microbial taxa of leaf were the bacterial species: Teredinibacter_sp._1162T.S.0a.05, a Rhizobiales ASV, a Alphaproteobacteria ASV, and a Dadabacteriales ASV (Fig. 5a). Bacterial species such as a rickettsiaceae ASV, a KI89A_clade ASV, and a Urania_IB_19_marine

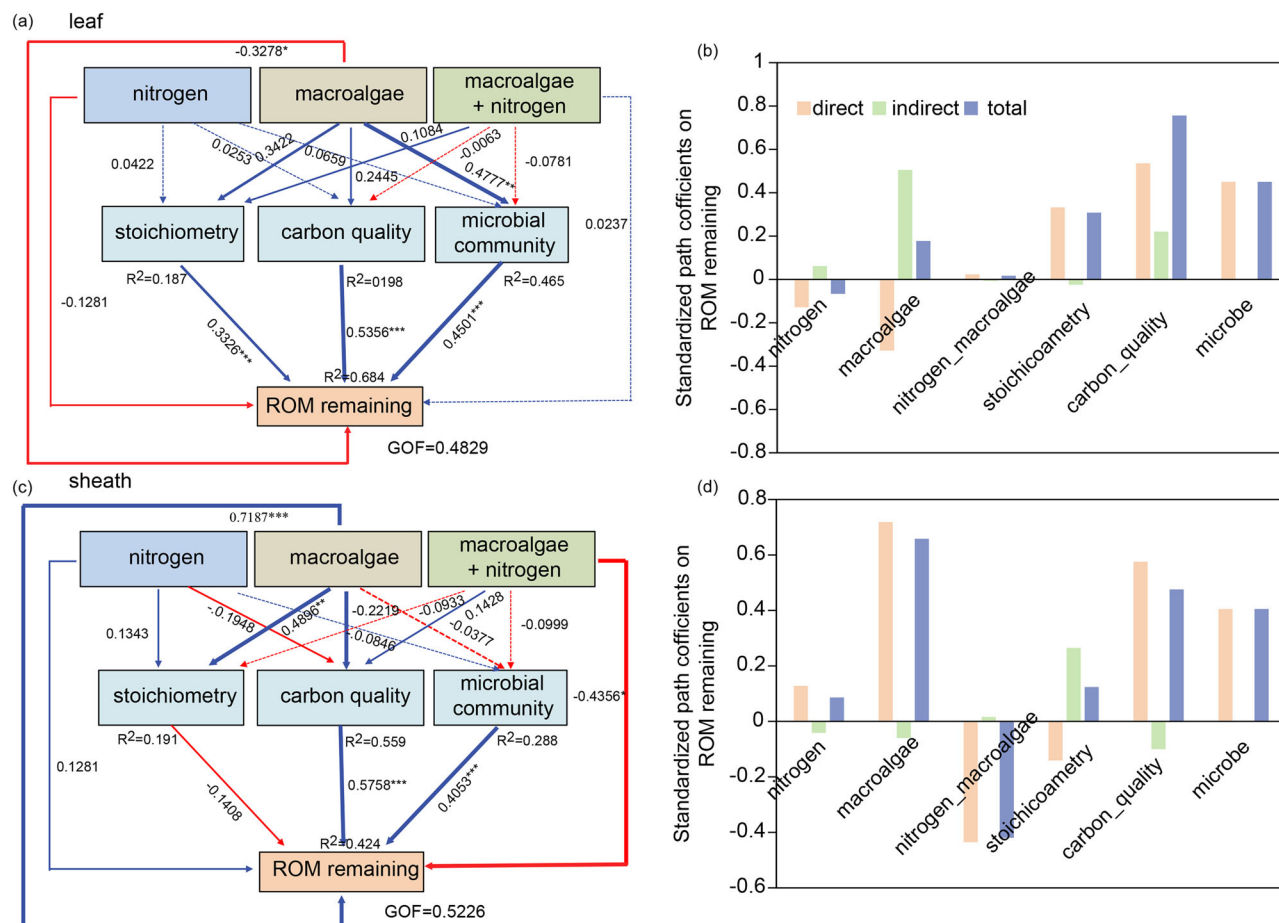


Fig. 4 | The relationships between N and macroalgae, biochemical properties (stoichiometry (C, N, C/N) and carbon quality (TI₁, TI₂, TI₃) of litter, and bacterial and fungal community characteristics (bacterial and fungal ASV richness). a, c Final results of partial least squares path model (PLS-PM) analysis of the effects of N and macroalgae treatments on leaf and sheath. ROM remaining refers to the sum of the remaining proportions of litter TI₂ and TI₃. The standardized path coefficients were shown near arrows. The value of each path coefficient is indicated by the arrow width; blue and red arrows indicate positive and negative effects,

respectively. Values of R^2 represent the explanation rates contributed by variables that point to it. **b, d** The standardized path coefficients for direct and indirect effects on the ROM remaining. Different nitrogen levels: no nitrogen addition (Control, Macroalgae); low nitrogen addition (Low N, Macroalgae + Low N); high nitrogen addition (High N, Macroalgae + High N), $n = 6$ each. Different macroalgae levels: no macroalgae addition (Control, Low N, High N); macroalgae addition (Macroalgae, Macroalgae + Low N, Macroalgae + High N), $n = 9$ each.

_sediment_group ASV played a key role in sheath ROM decomposition (Fig. 5b). The performance of models was included (Supplementary Table 16).

Discussion

Our findings demonstrate that organic matter composition and microbial community structure exhibit temporal dynamic shifts during the decomposition of seagrass litter. The nitrogen dynamics in leaves except for the treatments without macroalgae addition follow a distinct 'leaching-immobilization-release' pattern, whereas the nitrogen dynamics in sheaths exhibit greater complexity. The increase in the remaining proportion of nitrogen of sheaths occurs earlier than that of leaves. It has been demonstrated that litter with a C:N > 30 tends to accumulate these elements^{35,36}. The sheath's initial C:N ratio was relatively high (>30), which explains why the sheath litter immobilized N at the early stage of decomposition³⁷. The leaf's C:N ratio was low (<30) during the decomposition process, which explains the constant decrease in treatments without macroalgae addition. During the decomposition period, the proportions of cellulose- and lignin-associated OM in leaf litter increased (Supplementary Fig. 9b, c). This increase may be due to a rapid decline in the percentage of labile organic matter (LOM) in the litter. The early stage of litter decomposition is characterized by the release of amounts of soluble materials and the degradation of unshielded cellulose³⁸, while late-stage decomposition typically involves

the degradation of lignin and lignified tissue³⁹. The bacterial communities in leaves and sheaths underwent a succession from *r*-strategists (Alphaproteobacteria, including Rhodobacterales and Rhizobiales)⁴⁰ to *K*-strategists (Chlamydiales and Microtrichales)^{41,42}, corresponding to the decomposition of labile and recalcitrant organic matter, respectively. The abundance of Saccharomycetales in sheaths decreased, while the abundance of Basidiomycetes gradually increased, consistent with previous study⁴³. However, no distinct succession of fungal communities was observed during the decomposition of leaf. This is likely due to rapid leaching shaping initial assembly. Despite declining labile carbon and accumulating recalcitrant compounds in later stages, minimal compositional shifts were observed due to habitat generalism (i.e., similar colonizing substrate)^{44,45}.

In partial agreement with hypothesis, we observed that the direction of microbial PE of seagrass leaves and sheaths and the corresponding microorganisms exhibit different patterns in the addition of N. The PLS-PM results showed that N addition facilitated leaf decomposition but hindered sheath decomposition. Specifically, N addition significantly accelerated the decomposition of cellulose-associated organic matter in leaf. The stimulation of cellulose decomposition is likely due to the increased relative abundance of microorganisms involved in cellulose degradation following N addition^{39,46}. In this study, N addition increased the relative abundance of Flavobacteriales (a Euzeyella ASV) and Actinobacteria (an Actinomarinales ASV) during leaf decomposition. Previous studies have demonstrated

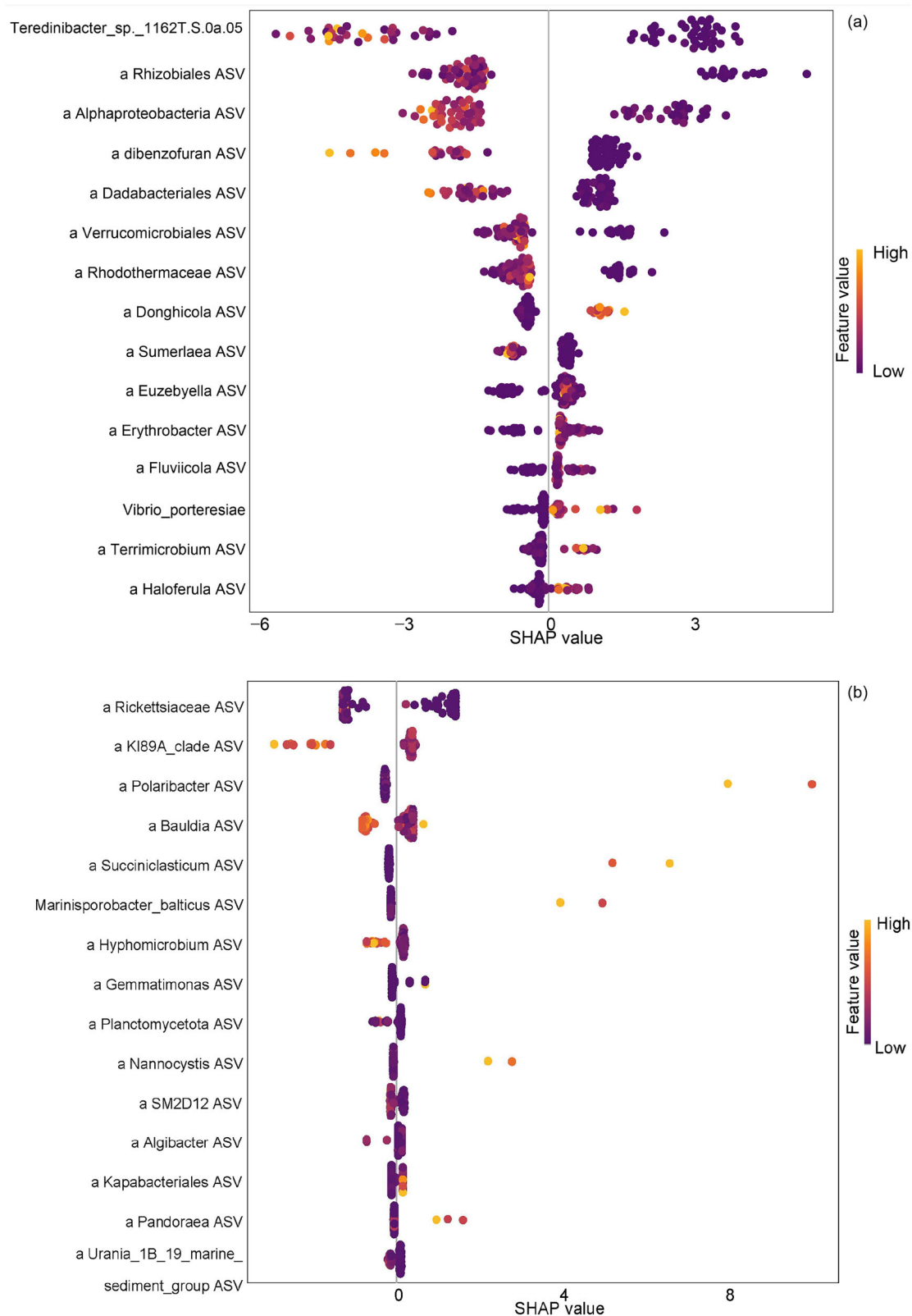


Fig. 5 | Contributions of input feature (bacteria) of the ROM remaining models. a Leaf; b sheath. The color bar indicates the low and high values of the features. The more widely spread the distribution of SHAP values for a feature, representing the greater influence of that feature on the model.

that Flavobacteriales and Actinobacteria play a key role in degrading litter recalcitrant polymers such as cellulose and lignin^{47–49}. RDA and SHAP analysis of leaf proved that these taxa promoted ROM decomposition. Thus, the presence of both *r*-strategists and *K*-strategists in bacteria likely accelerated leaf decomposition. In contrast to leaf, N addition decreased sheath

decay. The differing responses of leaf and sheath decomposition to N addition were related to litter quality, which further influenced changes in the bacterial community. In this study, the OM and N content of the sheath was lower than that of the leaf, whereas C/N, TI₃/OM, and TI₃/N ratios were higher (Supplementary Table 6). Further, the relative abundance of

Phycisphaerae (a *Urania_1B_19_marine_sediment_group* ASV) in sheath exhibited negative responses to N addition. Phycisphaerae has shown the potential to decompose complex carbon substrates⁵⁰, which may influence sheath decay. Moreover, studies have demonstrated that N may slow the breakdown of the litter with poor quality (e.g., including high C/N or lignin/N) by inhibiting the synthesis of lignin-decomposing enzymes^{46,51}, which potentially explains the decline in sheath decay. β G facilitates cellulose degradation by hydrolyzing cellobiose, and its activity is commonly used to elucidate the mechanisms underlying priming effects⁵². Nitrogen addition elevated leaf β G activity but reduced sheath β G activity, demonstrating stimulated decomposition of recalcitrant organic matter in leaves and suppressed decomposition in sheaths. These opposing effects jointly influence the carbon balance of seagrass beds, thus requiring a comprehensive assessment of their net impact on carbon release and sequestration.

Macroalgae addition caused changes in biochemical properties of the litter (i.e., litter mass, C, N, LOM, cellulose, and lignin-associated OM) and the microbial community structure, ultimately affecting the decomposition of seagrass leaves and sheaths. The addition of macroalgae decreased the mass loss of leaves and sheaths by 12% and 21%, respectively. Leaf and sheath N remaining increased during the rapid decomposition periods of macroalgae, partly due to an augmented microbial immobilization process⁵³ or nutrient transfer in mixed litter⁵⁴. Mixed litter provides a diverse range of substrates and ecological niches for microorganisms, promoting synergistic interactions among different microbial communities, thereby facilitating nitrogen immobilization⁵³. Meanwhile, nutrient transfer, which refers to nutrients (such as N and P), is transferred from high-quality litter to low-quality litter during the decomposition process of mixed litter⁵⁴. The initial N content of macroalgae was higher than that of leaves or sheaths in this study. Over time, the N remaining of leaves and sheaths increased while that of macroalgae decreased (Supplementary Fig. 2a, e and Supplementary Fig. 10a, b), supporting the nutrient transfer from macroalgae to leaf and sheaths. Macroalgae addition resulted the negative PE through impeding the loss of both LOM and ROM in the leaf and sheath. Further, the PLS-PM results indicated that macroalgae addition inhibited the decomposition of ROM in the leaf and sheath. In this study, the presence of macroalgae in leaf and sheath increased the abundance of Proteobacteria (a *Ruegeria* ASV and a *Woeseia* ASV) and Microtrichales (a *Sva0996_marine_group* ASV), while decreased the abundance of a Rhizobiales ASV, a Chlamydiales ASV, and a Dadabacteriales ASV. Notably, Rhizobiales, which are known for their role in N fixation and organic carbon PE^{55,56}. RDA and SHAP analysis in this study also indicated that Rhizobiales were important to the decay of OM. Fungi play a crucial role in decomposing ROM in seagrass meadows as well⁵⁷. In this study, the addition of macroalgae to leaf litter decreased the relative abundance of Sordariomycetes (*Lecythophora_luteoviridis* and a *Fusarium* ASV), Dothideomycetes (*Tyrannosorus_hystrionoides*), Leotiomycetes (*Pseudeurotium_ovale*), and Agaricomycetes (*Psathyrella_lutensis*), all of which are known to be involved in lignocellulose degradation^{58–61}. It has been reported that the preferential substrate utilization is the key mechanism of negative PE, which refers to that microorganisms may shift their focus from decomposing litter to utilizing exogenous labile organic carbon, neglecting the ROM of the litter⁶². Further, study has shown that under conditions of abundant C and N surpluses, litter-decomposing *K*-strategists (oligotrophs) tend to shift their utilization towards labile organic carbon, leading to a real negative PE⁶³. This study demonstrates that macroalgae addition decreased β G activity in the sheath. Macroalgae decomposition releases abundant readily degradable carbon sources (e.g., soluble sugar), which are rapidly utilized by microorganisms, thereby downregulating of β G synthesis and secretion⁶⁴. For decay of leaf and sheath in this study, we postulated that *r*-strategists and *K*-strategists in bacteria taxa turned to colonize macroalgae substrate, an exogenous source of labile organic carbon.

The influence of combined N and macroalgae addition on the decomposition of recalcitrant ROM in seagrass litter also varied with tissue types. Both the repeated measures analysis of variance and PLS-PM results indicated that the interactive effects of macroalgae and N on leaf

decomposition, particularly ROM decomposition, were minimal. This could be attributed to the intricate competition between *r*-strategists and *K*-strategists, with the latter favoring growth in nutrient-poor environments⁶⁵. The relative abundance of bacteria such as Microtrichales (a *Sva0996_marine_group* ASV) was lower in combined macroalgae and N addition than that in macroalgae alone, while the relative abundance of Flavobacteriales (a *Arenibacter* ASV and a *Euzebyella* ASV) and Alphaproteobacteria (*Cohaesibacter* and *Thioclava_pacifica*) showed the opposite trends. Elevated nutrient inputs may favor the proliferation of *r*-strategists in bacteria, while inhibiting the growth of *K*-strategists in bacteria. The complex microbial interactions led to no significant additive effect of N and macroalgae on ROM decomposition in leaves. Conversely, the interactive effects of nitrogen and macroalgae significantly influence sheath mass (measured at 37 and 116 days) and organic matter remaining (TI₂ and TI₃). Macroalgae and N exhibited an antagonistic interaction, where the macroalgae effect was nitrogen-dependent: under natural nitrogen levels, macroalgae significantly inhibited sheath decomposition, but this inhibitory effect was attenuated with increasing nitrogen availability. For bacteria, Actinomarinales (an Actinomarinales ASV) and Gammaproteobacteria (a *Candidatus_Berkiella* ASV) were more abundant under N combined with macroalgae addition, potentially driving the loss of sheath mass and ROM. A study has shown that Gammaproteobacteria are putative *K*-strategists (oligotrophs)⁶⁶. We conclude that the addition of macroalgae combined with N may mitigate the negative PE of macroalgae addition by providing ample N resources for microorganisms.

The amplified nutrient load from coastal eutrophication diminishes the production of seagrass biomass, thereby reducing its carbon (C) storage capacity⁶⁷. In this study, we similarly found that eutrophication enhances seagrass biomass decomposition, further reducing C storage capacity of these climate-regulating ecosystems. The decay-enhancing impact of eutrophication was primarily due to either (a) inorganic N availability that increased recalcitrant C decomposition in the leaves or (b) the increase in sheath carbon decay with the combined addition of N and macroalgal biomass. Given that recalcitrant or refractory OM accounts for ~50% of seagrass above-ground litter biomass²⁹ and sheaths can persist for millennia within a mat⁶⁸, these decomposition dynamics have profound implications for long-term carbon sequestration. In contrast to the effects of N addition, we discovered that the addition of green macroalgae *Ulva* did not promote litter decomposition. Instead, macroalgae addition inhibited seagrass ROM loss, resulting in a negative PE and a pathway for reduced ROM mineralization. This phenomenon may arise from microbial metabolic shifts: under abundant labile organic matter (e.g., macroalgae), decomposers prioritize low-energy degradation pathways for readily available substrates, downregulating energy-intensive enzymes like laccase that target recalcitrant compounds^{69,70}. Although macroalgal blooms proliferate under favorable climatic conditions (e.g., high sea surface temperature and atmospheric CO₂) rather than exclusively under the strain of high nutrient loading⁷¹, their demonstrated inhibitory effects on litter decomposition rates neither result in enhanced carbon sequestration within seagrass bed ecosystems nor promote the accumulation of ROM. Such a paradox arises because macroalgal blooms are well-documented to decrease seagrass biomass through light shading²³. Future studies should comprehensively evaluate the impacts of macroalgal blooms by integrating both decomposition rates and litter biomass dynamics.

A comprehensive review revealed a spectrum of algal-mediated PE, ranging from strongly negative to strongly positive⁵³. Further, Guenet et al.⁷² suggested that the PE switch from a positive PE in oligotrophic systems to a negative PE in eutrophic conditions. Over-mineralization of ROM may largely contribute to the CO₂ emissions of aquatic ecosystems through the positive PE⁷². In contrast to the findings in our study, *Z. muelleri* mixed with *U. lactuca* (initially averaging 1.31% N and 0.80% N, respectively) promoted *Z. muelleri* decomposition through a co-metabolism effect, indicative of a positive PE⁷. The N contents of *T. hemprichii* and *U. pertusa* in this study were relatively higher (initially averaging 2.62% N and 3.47% N, respectively). These contrasting results show that the quality (e.g., N-availability)

of lignocellulose-containing seagrass biomass is an important factor influencing the direction, and perhaps magnitude of PE in seagrass ecosystems. High-quality litter (e.g., high N content) inputs can boost microbial carbon use efficiency and reduce the magnitude of priming effect through providing adequate nitrogen resources. Conversely, low-quality litter (e.g., low N content) elevated nutrient demands of microorganisms, which drives them to mine nitrogen from existing organic matter⁷³. Leaf N has been recognized as a metric of chronic eutrophication⁷⁴. Therefore, we recommend conducting mixed decomposition experiments involving seagrass litter and macroalgae with varying initial N or P concentrations to elucidate the relationship between plant tissue quality and PE. By identifying the litter quality threshold of PE and linking litter nitrogen to external nutrient load levels, it is crucial for predicting the response patterns of carbon cycling in coastal ecosystems to eutrophication stress. This approach not only reveals the mechanisms by which nutrient inputs influence carbon dynamics but also provides a scientific foundation for developing targeted management strategies to effectively mitigate the adverse impacts of eutrophication on coastal ecosystems.

Conclusion

Our study found that different rates of decomposition of seagrass leaves and sheaths may be related to differences in their initial nutrient composition and carbon composition. These differences trigger changes in the microbial community and the directions of priming effect on seagrass litter decomposition under eutrophication conditions. Macroalgal blooms inhibited the decomposition of litter with contrasting quality. Nutrient thresholds likely mediate the impacts of macroalgal blooms on seagrass litter carbon pools. The interactive effects of nitrogen loading and macroalgal blooms barely influence high-quality leaf decomposition, while they enhance that of low-quality sheaths. Consequently, litter quality and microbes must be taken into account when assessing the effect of eutrophication on decomposition. Looking ahead, global climate change may worsen eutrophication, exposing seagrass ecosystems to more complex environmental conditions regarding litter decomposition. Considering the identified effects and climate scenarios, differentiated nutrient control strategies are recommended: prioritizing conservation of high-quality seagrass tissues in high nitrogen loading areas and establishing dynamic monitoring and early-warning systems in macroalgal bloom-prone regions. Overall, our results underscore the necessity for regular monitoring nutrient loading to facilitate effective conservation of seagrass ecosystems and strengthen their role as blue carbon sinks in the context of global climate governance.

Materials and methods

Material collection and preparation

The research was conducted in Li'an lagoon, located on Hainan Island in southeast China (41°51'N, 124°54'E). Li'an is a nearly enclosed natural lagoon with only one opening connecting it to the open sea. The seagrass meadow in Li'an has an area of about 320 ha and a water depth of 0–3 m⁷⁵. This marine ecosystem is influenced by the diurnal tidal cycles. Owing to substantial inflows of domestic sewage and aquaculture wastewater, the water quality in Li'an is poor (e.g., high nutrient loading), resulting in widespread macroalgal blooms that blanket the seagrass in the intertidal area. Macroalgae, notably *U. pertusa*, experience seasonal surges, whereas *T. hemprichii* is consistently the dominant seagrass species.

In September 2021, living *T. hemprichii* and *U. pertusa* were gathered from Li'an. Utilizing fresh biomass for the experiment, rather than beach wrack, ensured uniformity in age and avoided extensive decomposition of the starting material. Both *T. hemprichii* and *U. pertusa* were collected within 0.25 m² quadrats at the sampling site during low tide, then manually removed the epiphytes on the seagrass leaf, cleaned with Milli-Q water, dried at 60 °C for 48 h, and weighed to determine macroalgal and seagrass aboveground biomass. Additionally, a total of ~5 kg and 10 kg of fresh *T. hemprichii* and *U. pertusa* were collected from sampling sites, respectively. Seawater and surface sediment (coarse sand; ~5 cm deep) were collected on-site, and all materials were promptly transported to the

laboratory. The seagrass and macroalgae underwent rinsing with Milli-Q water before subsampling. After removing all visible invertebrates and plant litter, the collected sediment was homogenized and distributed into six microcosms, each measuring 23 × 23 × 30 cm. ~5 cm of sediment was added to each microcosm, which was then filled with the collected seawater to a depth of ~20 cm.

Experimental design and incubation study

Six experimental groups were created: seagrass (Control), seagrass with low NH₄⁺ (Low N), seagrass with high NH₄⁺ (High N), seagrass combined with macroalgae (Macroalgae), seagrass and macroalgae with low NH₄⁺ addition (Macroalgae+ Low N), and seagrass and macroalgae with high NH₄⁺ addition (Macroalgae+ High N). N was introduced into the water column of mesocosms in the form of NH₄Cl (1 mol L⁻¹), resulting in NH₄⁺ concentrations of 20–30 μM for the low NH₄⁺ addition treatment and 40–50 μM for the high NH₄⁺ addition treatment, while the control group received no NH₄Cl, with a natural NH₄⁺ concentration of 0.44 μM. The schematic diagram of the experimental design is shown in Supplementary Fig. 11. The range of nitrogen concentration was set based on the reported in situ NH₄⁺ concentrations in seawater of seagrass beds^{76,77}. The actual NH₄⁺ concentrations in NH₄Cl-amended treatments were as follows: Low N: 24.12 ± 0.05 μM, High N: 41.43 ± 3.31 μM, Macroalgae+ Low N: 23.92 ± 1.15 μM, Macroalgae+ High N: 46.96 ± 3.49 μM. This experiment adopted a 2 × 3 factorial design, whereas previous studies only considered the individual effects of macroalgae or nutrients alone and employed limited nutrient gradients^{7,19}.

Before commencing the experiment, entire leaves and sheaths were sorted from the seagrass plants by hand. The leaves are strap-shaped, slightly curved like a sickle. The sheath, located at the leaf base, envelops the stem and remains attached to it. These tissues can be distinguished by their distinct colors and forms. The whole leaf was used, but those with obvious signs of herbivore attack and senescence were excluded. To maintain the consistency of the material used for the decay experiment, large epiphytes were manually removed, but no scraping was performed to prevent damage to the leaves⁷⁸. Leaves and sheaths were trimmed to lengths of 2–3 cm to ensure homogeneity of initial decomposition substrates according to previous studies^{7,79}, while intact thalli of macroalgae (5–7 cm in length) were used due to their fragility. Litter was then thoroughly mixed to provide a homogeneous pool of each tissue type. The mixed litter was dried with filter paper to remove excess water⁸⁰. Aliquots (leaves: ca. 4.5–5 g; sheath: ca. 3 g wet weight) of each fresh litter material were weighed and sealed in separate litterbags (20 × 15 cm, mesh size 1 mm) for Control, Low N, and High N treatments. Based on the calculated biomass data, a mixture of seagrass and macroalgae (1:3 mass wet weight ratio) was enclosed in litterbags for Macroalgae, Macroalgae+ Low N, and Macroalgae+ High N treatments. The mesh size of the litterbags was chosen to minimize litter loss from mechanical fragmentation and limit the macrofauna on decomposition⁸¹. Simultaneously, numbered insert labels were placed inside the litterbags.

All 144 litterbags (6 treatments × 4 sampling times × 3 replicates × 2 seagrass tissues) were evenly divided into six groups, with 24 litterbags placed in each mesocosm. The litterbags were positioned at the surface of the seawater in a floating state to simulate the decomposition of seagrass aboveground tissues in natural environments. The experiment was conducted in the dark to prevent photodegradation. To maintain a constant salinity and water depth of 20 cm throughout the incubation period, Milli-Q water was periodically supplemented to compensate for evaporative losses. Simultaneously, dissolved oxygen concentrations in the seawater were consistently maintained through continuous aeration using an air pump system. One aeration tube was deployed at identical positions across all mesocosms.

Each treatment of 2 different seagrass tissues was sampled at days 15, 37, 65, and 116, with 3 replicates per sampling time. These time points were selected to capture the rapid leaching phase of litter (typically occurring within the initial 1–2 weeks of decomposition) and the subsequent prolonged microbial colonization and stabilization phases (37–116 days)^{82,83}. The decomposition period of 116 days encompassed three decomposition

phases (i.e., leaching, microbial colonization, stabilization) of seagrass aboveground tissues and macroalgae^{7,84}, with minimal mass loss observed in the litter after microbial colonization process⁸⁵. Therefore, the incubation period of 116 days was scientifically justified to elucidate phase-specific decomposition dynamics. At each sampling time, three bags were randomly collected from seagrass leaves or sheaths of each treatment, where the samples were fauna eliminated. The litter was extracted and sorted into leaf, sheath, and macroalgal components. During decomposition, litter can be distinguished based on color and morphology. While macroalgal litter remains green during decomposition, leaf and sheath exhibit brown pigmentation. Notably, leaf transition from green to brown due to rapid leaching of labile organic compounds and subsequent microbial colonization. A small portion of the samples (seagrass leaves and sheaths) of all treatments was dried with filter paper and weighed, placed into Eppendorf tubes (2 ml), and then stored in a freezer at -80°C for subsequent microbial analysis. For enzyme assays, additional litter samples were weighed and placed in 2 mL Eppendorf tubes and immediately refrigerated at 4°C . The remaining samples were also weighed and then dried at 60°C for 48 h, and weighed dry weight (DW), and the percentage of mass remaining at each sampling time was calculated.

Three additional ca. 2 g litter samples of each seagrass tissue (i.e., leaves and sheaths) and macroalgae *U. pertusa* were also dried, and the initial DW of the litter used for incubation was corrected according to the ratio of wet to oven-dried weight. Then, the dried samples were used to determine initial biochemical components (details are shown in Supplementary Table 6).

C, N, total phenol, and TGA analysis

The dried litter samples of *T. hemprichii* and *U. pertusa* were homogenized with a grinder instrument (mortar grinder ST-B220, SKTEM, China). Subsequently, the C and N content of samples (sample quantity: 1.5–2.5 mg DW) was measured using an elemental analyzer (Vario EL, Elemental Analyser System GmbH, Germany). Total phenol content (sample quantity: 0.02–0.03 g DW) in the samples was determined using the Folin-Ciocalteu reagent⁸⁶.

The remaining litter samples of *T. hemprichii* underwent thermogravimetric analysis (TGA) using a TG209F1 (Netzsch, Germany) instrument with a sensitivity of $0.1\text{ }\mu\text{g}$. Approximately 10 mg of the ground sample was placed in a platinum cup and heated under N_2 (gas flow = 100 mL min^{-1}) at a ramp rate of $10^{\circ}\text{C min}^{-1}$, ranging from 35 to 650°C ³³. The thermal intervals (TI) are described by the rate-of-change derivative (% mass loss $^{\circ}\text{C}^{-1}$). Three temperature thermal intervals (TI₁, TI₂, and TI₃) were selected according to the organic matter lability and the percentage of mass change. TI₁, representing labile components such as soluble carbohydrates and hemicellulose, spanned from 180°C to 300°C ^{9,33}. TI₂, denoting cellulose-associated recalcitrant OM, ranged from 300°C to 400°C , while TI₃, encompassing lignin-associated recalcitrant OM and insoluble polysaccharide residues, extended from 400°C to 600°C ^{29,33}. Mass loss above 600°C mainly represents inorganic substances, such as sulfur residues^{29,33}. The range of mass loss (TI₁₋₃) was recalculated as a portion of total organic matter from 180°C to 600°C ²⁹. The proportions of TI₁, TI₂, and TI₃ in total OM were represented as TI₁/OM, TI₂/OM, and TI₃/OM²⁹.

Microbial community and enzyme activity analyses

The litter samples stored in Eppendorf tubes were used for a targeted analysis of the seagrass-associated bacterial and fungi community. According to the variations of seagrass litter qualities (based on biomass, elemental content, and TGA), we chose to use seagrass leaf and sheath litter over the whole decomposition process. Bacterial community employed 16S rRNA amplicon sequencing and DADA2 bioinformatics pipeline. Fungal community analysis was based on ITS1 amplicon sequencing and the DADA2 bioinformatics pipeline (seen details in the Supplementary Methods). Determination of β -glucosidase activity in litter was described in Supplementary Methods.

Statistical analysis

The decomposition rate of mass and OM (TI₁, TI₂, TI₃) in leaf and sheath litter was determined using the Olson exponent model. This model was conducted in Origin 9.0. The Olson exponent model assumes that litter is homogeneous and that the decomposition coefficients (k) remain unchanged throughout the process⁸⁷. While this model overlooks the variability in litter components, its simplicity and ease of application have made it widely used in decomposition studies^{19,88}. Alternative decomposition models, such as the double or triple-component exponential model, are more appropriate for prolonged decomposition periods exceeding one year⁸⁸ and hence were not employed in this study. Details on calculating the decomposition coefficients, the time needed for 50% ($T_{50\%}$) and 95% ($T_{95\%}$) decomposition of mass, as well as the remaining proportions of elements, OM (TI₁, TI₂, TI₃), ROM (sum of the proportions of TI₂ and TI₃), and total phenol, were outlined in the Supplementary Methods. One-way ANOVA and Tukey post-hoc test were used to analyze differences in the biochemical components (C, N, C/N, TI₁, TI₂, TI₃, TI₁/OM, TI₂/OM, TI₃/OM, TI₃/N, and total phenol) of leaves, sheaths, and macroalgae before the decomposition process. The TI₃/N ratio corresponds to the lignin-to-nitrogen ratio (lignin/N), a widely recognized indicator for evaluating litter decomposition rates and subsequent nutrient release patterns⁸⁹. Repeated measures analysis of variance (ANOVA) was conducted to evaluate the effects of N and macroalgae additions on litter biochemical compositions (e.g., remaining proportions of mass, C, N, C/N, TI₁, TI₂, TI₃, and total phenol), and Alpha diversity indexes of microbial communities (ASV richness) of litter. Additionally, a two-way ANOVA was used to analyze the difference in these parameters across various decay times and the enzyme activity in litter at 37 days. The model included nitrogen (N, with three levels: no nitrogen addition, Low N, and High N) and macroalgae (M, with two levels: with macroalgae and without macroalgae) as fixed factors. For analyzing the effects of the nitrogen factor on litter biochemical compositions and amplicon sequence variant (ASV) richness, the Tukey post-hoc test was applied for multiple comparisons. Meanwhile, the macroalgae factor had two levels, and an independent t -test was used to compare the difference between the groups. Simple effects analysis and pairwise comparisons (Bonferroni adjustment) were used for interactions of nitrogen and macroalgae. Before conducting one-way ANOVA, repeated measures ANOVA, and two-way ANOVA tests, we performed the Mauchly and Levene's test for sphericity and homogeneity of variance, respectively. Data were log-transformed to achieve the homogeneity of variance where the heterogeneity of variance was found. Mass remaining, C, N, TI₁₋₃, TI₁/OM, TI₂/OM, and TI₃/OM were converted into arcsine square roots before ANOVA. One-way ANOVA, repeated measures analysis of variance, two-way ANOVA, Tukey post-hoc test, and independent t -test were conducted in SPSS 23.0. Simple effects analysis was conducted using the emmeans package in R software version 4.4.1.

Redundancy analysis (RDA) was performed to investigate the relationships between microbial species and litter biochemical compositions. Two-dimensional nonmetric multidimensional scaling (NMDS) analyses of litter microbial communities were performed based on Bray-Curtis distances. Permutational analysis of variance (PERMANOVA) and permutational test for homogeneity of multivariate dispersions (PERMDISP) utilizing the Bray-Curtis distance matrix were employed to identify differences in litter bacterial and fungal communities among N and macroalgae addition treatments⁹⁰. The similarity percentage (SIMPER) analysis, coupled with Wilcoxon signed-rank tests, was utilized to identify the microbial taxa contributing the most to the dissimilarities in community composition between treatments (abundance values were square root transformed). RDA, NMDS, PERMANOVA, PERMDISP, and SIMPER analyses were carried out using the vegan package in R software version 4.4.1.

A multi-group Partial Least Squares Path Model (PLS-PM) was used to explore the effect of N and macroalgae addition on the stoichiometry and carbon quality of litter, and bacterial and fungal community characteristics. Carbon quality refers to the chemical complexity of organic carbon in litter, encompassing labile compounds (e.g., soluble sugars) and recalcitrant

components such as lignocellulose⁹¹. PLS-PM is a particularly useful statistical method for demonstrating cause and effect relationships among exogenous variables and observed variables⁹². A random forest model and shapley additive explanations (SHAP) analysis were performed to identify key microbial taxa at the species level controlling for ROM decomposition of litter^{93,94}. This analysis was also carried out using R software (details seen in the Supplementary Methods).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data sets generated in this study are available at Figshare: primary datasets under <https://doi.org/10.6084/m9.figshare.29291471.v2>, and supplementary Tables 2, 4, 7, 8, 9, 10, 12, 13, and 15 under <https://doi.org/10.6084/m9.figshare.29432387>. The raw sequencing data are deposited in the NCBI BioProject database under accession number PRJNA1276285.

Received: 30 December 2024; Accepted: 4 July 2025;

Published online: 17 July 2025

References

- Duarte, C. M. & Chiscano, C. L. Seagrass biomass and production: a reassessment. *Aquat. Bot.* **65**, 159–174 (1999).
- Duarte, C. M., Middelburg, J. J. & Caraco, N. Major role of marine vegetation on the oceanic carbon cycle. *Biogeosciences* **2**, 1–8 (2005).
- Kennedy, H. et al. Seagrass sediments as a global carbon sink: isotopic constraints. *Global Biogeochem. Cycles* **24**, GB4026 (2010).
- Lavery, P. S., McMahon, K., Weyers, J., Boyce, M. C. & Oldham, C. E. Release of dissolved organic carbon from seagrass wrack and its implications for trophic connectivity. *Mar. Ecol. Prog. Ser.* **494**, 121–133 (2013).
- Boudouresque, C. F. et al. The necromass of the *Posidonia oceanica* seagrass meadow: fate, role, ecosystem services and vulnerability. *Hydrobiologia* **781**, 25–42 (2016).
- Duarte, C. M. & Cebrian, J. The fate of marine autotrophic production. *Limnol. Oceanogr.* **41**, 1758–1766 (1996).
- Liu, S. L., Trevathan-Tackett, S. M., Lewis, C. J. E., Huang, X. P. & Macreadie, P. I. Macroalgal blooms trigger the breakdown of seagrass blue carbon. *Environ. Sci. Technol.* **54**, 14750–14760 (2020).
- Romero, J., Pergent, G., Pergentmartini, C., Mateo, M. A. & Regnier, C. The detritic compartment in a *Posidonia oceanica* meadow: litter features, decomposition rates, and mineral stocks. *Mar. Ecol.* **13**, 69–83 (1992).
- Liu, S. L. et al. Nutrient loading weakens seagrass blue carbon potential by stimulating seagrass detritus carbon emission. *Ecol. Indic.* **157**, 10 (2023).
- Ouyang, X. et al. Response of macrophyte litter decomposition in global blue carbon ecosystems to climate change. *Global Change Biol.* **29**, 3806–3820 (2023).
- Qin, L.-Z. et al. Macrophyte litter mixtures mediate decomposition processes in coastal sediments. *Commun. Earth Environ.* **5**, 610 (2024).
- Orth, R. J. et al. A global crisis for seagrass ecosystems. *Bioscience* **56**, 987–996 (2006).
- Rabalais, N. N., Turner, R. E., Diaz, R. J. & Justic, D. Global change and eutrophication of coastal waters. *Ices J. Mar. Sci.* **66**, 1528–1537 (2009).
- Waycott, M. et al. Accelerating loss of seagrasses across the globe threatens coastal ecosystems. *Proc. Natl. Acad. Sci. USA* **106**, 12377–12381 (2009).
- Li, M., Lundquist, C. J., Pilditch, C. A., Rees, T. A. V. & Ellis, J. Implications of nutrient enrichment for the conservation and management of seagrass *Zostera muelleri* meadows. *Aquat. Conserv.* **29**, 1484–1502 (2019).
- VanKatwijk, M. M., Vergeer, L. H. T., Schmitz, G. H. W. & Roelofs, J. G. M. Ammonium toxicity in eelgrass *Zostera marina*. *Mar. Ecol. Prog. Ser.* **157**, 159–173 (1997).
- Villazan, B., Salo, T., Brun, F. G., Vergara, J. J. & Pedersen, M. F. High ammonium availability amplifies the adverse effect of low salinity on eelgrass *Zostera marina*. *Mar. Ecol. Prog. Ser.* **536**, 149–162 (2015).
- Burkholder, J. M., Tomasko, D. A. & Touchette, B. W. Seagrasses and eutrophication. *J. Exp. Mar. Biol. Ecol.* **350**, 46–72 (2007).
- Apostolaki, E. T., Marba, N., Holmer, M. & Karakassis, I. Fish farming impact on decomposition of *Posidonia oceanica* litter. *J. Exp. Mar. Biol.* **369**, 58–64 (2009).
- Qin, L.-Z., Suonan, Z., Kim, S. H. & Lee, K.-S. Coastal sediment nutrient enrichment alters seagrass blue carbon sink capacity. *Environ. Sci. Technol.* **55**, 15466–15475 (2021).
- Hobbie, S. E. et al. Response of decomposing litter and its microbial community to multiple forms of nitrogen enrichment. *Ecol. Monogr.* **82**, 389–405 (2012).
- Knorr, M., Frey, S. D. & Curtis, P. S. Nitrogen additions and litter decomposition: a meta-analysis. *Ecology* **86**, 3252–3257 (2005).
- McGlashery, K. J. Macroalgal blooms contribute to the decline of seagrass in nutrient-enriched coastal waters. *J. Phycol.* **37**, 453–456 (2001).
- Brewton, R. A. & Lapointe, B. E. The green macroalga *Caulerpa prolifera* replaces seagrass in a nitrogen enriched, phosphorus limited, urbanized estuary. *Ecol. Indic.* **156**, 111035 (2023).
- Kuzyakov, Y., Friedel, J. K. & Stahr, K. Review of mechanisms and quantification of priming effects. *Soil Biol. Biochem.* **32**, 1485–1498 (2000).
- Zhang, Z. et al. Priming effects of soil organic matter decomposition with addition of different carbon substrates. *J. Soil. Sediment.* **19**, 1171–1178 (2019).
- Trevathan-Tackett, S. M., Thomson, A. C. G., Ralph, P. J. & Macreadie, P. I. Fresh carbon inputs to seagrass sediments induce variable microbial priming responses. *Sci. Total Environ.* **621**, 663–669 (2018).
- Thomson, A. C. G., Trevathan-Tackett, S. M., Maher, D. T., Ralph, P. J. & Macreadie, P. I. Bioturbator-stimulated loss of seagrass sediment carbon stocks. *Limnol. Oceanogr.* **64**, 342–356 (2019).
- Trevathan-Tackett, S. M. et al. A global assessment of the chemical recalcitrance of seagrass tissues: implications for long-term carbon sequestration. *Front. Plant Sci.* **8**, 925 (2017).
- Fanin, N., Alavoine, G. & Bertrand, I. Temporal dynamics of litter quality, soil properties and microbial strategies as main drivers of the priming effect. *Geoderma* **377**, 114576 (2020).
- Chao, L. et al. Litter carbon and nutrient chemistry control the magnitude of soil priming effect. *Funct. Ecol.* **33**, 876–888 (2019).
- Kaal, J. et al. Molecular composition of plant parts and sediment organic matter in a Mediterranean seagrass (*Posidonia oceanica*) mat. *Aquat. Bot.* **133**, 50–61 (2016).
- Trevathan-Tackett, S. M. et al. Comparison of marine macrophytes for their contributions to blue carbon sequestration. *Ecology* **96**, 3043–3057 (2015).
- Mukai, H. Biogeography of the tropical seagrasses in the western Pacific. *Mar. Freshwater Res.* **44**, 1–17 (1993).
- Cai, A. et al. Patterns and driving factors of litter decomposition across Chinese terrestrial ecosystems. *J. Clean. Prod.* **278**, 123964 (2021).
- Li, Z., Peng, Q., Dong, Y. & Guo, Y. The influence of increased precipitation and nitrogen deposition on the litter decomposition and soil microbial community structure in a semiarid grassland. *Sci. Total Environ.* **844**, 157115 (2022).
- Bargali, S. S., Shukla, K., Singh, L., Ghosh, L. & Lakhera, M. L. Leaf litter decomposition and nutrient dynamics in four tree species of dry deciduous forest. *Trop. Ecol.* **56**, 191–200 (2015).

38. Yue, K. et al. Degradation of lignin and cellulose during foliar litter decomposition in an alpine forest river. *Ecosphere* **7**, e01523 (2016).
39. Berg B., McClaugherty C., Berg B. & McClaugherty C. *Plant Litter: Decomposition, humus formation, carbon sequestration*, 2nd edn (Springer Berlin Heidelberg, 2008).
40. Liu, S. et al. Nutrient loading diminishes the dissolved organic carbon drawdown capacity of seagrass ecosystems. *Sci. Total Environ.* **740**, 140185 (2020).
41. Wang, W. et al. Effects of mixed-species litter on bacterial and fungal lignocellulose degradation functions during litter decomposition. *Soil Biol. Biochem.* **141**, 107690 (2020).
42. Vadstein, O., Attramadal, K. J. K., Bakke, I. & Olsen, Y. K-selection as microbial community management strategy: a method for improved viability of larvae in aquaculture. *Front. Microbiol.* **9**, 2730 (2018).
43. Osono, T. Ecology of ligninolytic fungi associated with leaf litter decomposition. *Ecol. Res.* **22**, 955–974 (2007).
44. Ettinger, C. L. & Eisen, J. A. Fungi, bacteria and oomycota opportunistically isolated from the seagrass, *Zostera marina*. *PLoS ONE* **15**, e0251536 (2020).
45. Wainwright, B. J. et al. Seagrass-associated fungal communities show distance decay of similarity that has implications for seagrass management and restoration. *Ecol. Evol.* **9**, 11288–11297 (2019).
46. Carreiro, M. M., Sinsabaugh, R. L., Repert, D. A. & Parkhurst, D. F. Microbial enzyme shifts explain litter decay responses to simulated nitrogen deposition. *Ecology* **81**, 2359–2365 (2000).
47. Liu, G. et al. Climate warming enhances microbial network complexity by increasing bacterial diversity and fungal interaction strength in litter decomposition. *Sci. Total Environ.* **908**, 168444 (2024).
48. Stursová, M., Zifčáková, L., Leigh, M. B., Burgess, R. & Baldrian, P. Cellulose utilization in forest litter and soil: identification of bacterial and fungal decomposers. *FEMS Microbiol. Ecol.* **80**, 735–746 (2012).
49. Thi-Thu-Hong, L. et al. De novo metagenomic analysis of microbial community contributing in lignocellulose degradation in humus samples harvested from Cuc Phuong Tropical Forest in Vietnam. *Diversity* **14**, 220 (2022).
50. Lenferink, W. B. et al. Genomic analysis of the class Phycisphaerae reveals a versatile group of complex carbon-degrading bacteria. *Anton. Leeuw. Int. J. G.* **117**, 104 (2024).
51. Wu, A. et al. Forest gaps slow lignin and cellulose degradation of fir (*Abies faxoniana*) twig litter in an alpine forest. *Geoderma* **424**, 116010 (2022).
52. Li, J. et al. Microbial controls over soil priming effects under chronic nitrogen and phosphorus additions in subtropical forests. *ISME J* **17**, 2160–2168 (2023).
53. Halvorson, H. M., Francoeur, S. N., Findlay, R. H. & Kuehn, K. A. Algal-mediated priming effects on the ecological stoichiometry of leaf litter decomposition: a meta-analysis. *Front. Earth Sci.* **7**, 76 (2019).
54. Zhang, L., Zhang, Y., Zou, J. & Siemann, E. Decomposition of *Phragmites australis* litter retarded by invasive *Solidago canadensis* in mixtures: an antagonistic non-additive effect. *Sci. Rep.* **4**, 5488 (2014).
55. Wang, W., Chen, D., Zhang, Q., Sun, X. & Zhang, S. Effects of mixed coniferous and broad-leaved litter on bacterial and fungal nitrogen metabolism pathway during litter decomposition. *Plant Soil* **451**, 307–323 (2020).
56. Zhou, S., Wang, J., Chen, L., Wang, J. & Zhao, F. Microbial community structure and functional genes drive soil priming effect following afforestation. *Sci. Total Environ.* **825**, 153925 (2022).
57. Liu, S. et al. Nutrient loading decreases blue carbon by mediating fungi activities within seagrass meadows. *Environ. Res.* **212**, 113280 (2022).
58. Andersson, M., Kjoller, A. & Struwe, S. Microbial enzyme activities in leaf litter, humus and mineral soil layers of European forests. *Soil Biol. Biochem.* **36**, 1527–1537 (2004).
59. Yang, W. & Wang, K. Advances in forest soil enzymology. *Sci. Silv. Sin.* **40**, 152–159 (2004).
60. Schneider, T. et al. Who is who in litter decomposition? Metaproteomics reveals major microbial players and their biogeochemical functions. *ISME J* **6**, 1749–1762 (2012).
61. Kersten, P. & Cullen, D. Recent advances on the genomics of litter- and soil-inhabiting Agaricomycetes. (eds Horwitz, B. A., Mukherjee, P. K., Mukherjee, M., Kubicek, C. P.). (Springer Berlin, 2013).
62. Cheng, W. X. Rhizosphere feedbacks in elevated CO₂. *Tree Physiol.* **19**, 313–320 (1999).
63. Blagodatskaya, E. V., Blagodatsky, S. A., Anderson, T. H. & Kuzyakov, Y. Priming effects in Chernozem induced by glucose and N in relation to microbial growth strategies. *Appl. Soil Ecol.* **37**, 95–105 (2007).
64. Chen, H. et al. Effects of transcription factor CREA on the morphology and the secretion of β -glucosidase in *Aspergillus niger*. *Biotechnol. Bull.* **41**, 1–11 (2025).
65. Gul, S. & Whalen J. K. Perspectives and strategies to increase the microbial-derived soil organic matter that persists in agroecosystems. In: *Advances in Agronomy* (ed Sparks D. L.). (Academic Press, 2022).
66. Wang, Z. H. et al. N and P fertilization enhanced carbon decomposition function by shifting microbes towards an r-selected community in meadow grassland soils. *Ecol. Indic.* **132**, 108306 (2021).
67. Ravaglioli, C. et al. Macro-grazer herbivory regulates seagrass response to pulse and press nutrient loading. *Mar. Environ. Res.* **136**, 54–61 (2018).
68. Pergent, G. & Pergentmartini, C. Some applications of lepidochronological analysis in the seagrass *Posidonia-Oceanica*. *Bot. Mar.* **33**, 299–310 (1990).
69. Fontaine, S., Bardoux, G., Abbadie, L. & Mariotti, A. Carbon input to soil may decrease soil carbon content. *Ecol. Lett.* **7**, 314–320 (2004).
70. Von Luetzow, M. et al. Stabilization of organic matter in temperate soils: mechanisms and their relevance under different soil conditions - a review. *Eur. J. Soil Sci.* **57**, 426–445 (2006).
71. Feng, Y. et al. Shift in algal blooms from micro- to macroalgae around China with increasing eutrophication and climate change. *Global Change Biol.* **30**, e17018 (2024).
72. Guenet, B., Danger, M., Abbadie, L. & Lacroix, G. Priming effect: bridging the gap between terrestrial and aquatic ecology. *Ecology* **91**, 2850–2861 (2010).
73. Craine, J. M., Morrow, C. & Fierer, N. Microbial nitrogen limitation increases decomposition. *Ecology* **88**, 2105–2113 (2007).
74. Lee, K. S., Short, F. T. & Burdick, D. M. Development of a nutrient pollution indicator using the seagrass, *Zostera marina*, along nutrient gradients in three New England estuaries. *Aquat. Bot.* **78**, 197–216 (2004).
75. Huang, X. et al. Main seagrass beds and threats to their habitats in the coastal sea of South China. *Chin. Sci. Bull.* **51**, 136–142 (2006).
76. Ren, Y. Z. et al. Seagrass decline weakens sediment organic carbon stability. *Sci. Total Environ.* **937**, 173523 (2024).
77. Han, Q. et al. Characteristics of morphological and physiological indicators of *Thalassia hemprichii* and Environmental Factors in Xincun Bay, Hainan Island. *Prog. Fish. Sci.* **44**, 225–238 (2023).
78. Walker, D. I. & McComb, A. J. Decomposition of leaves from *Amphibolis-Antarctica* (Labill) Sonder et aschers and *Posdonia-Australis* Hook F the major seagrass species of Shark Bay, Western Australia. *Bot. Mar.* **28**, 407–413 (1985).
79. Zeng, J. et al. Differences in the regulation of soil carbon pool quality and stability by leaf-litter and root-litter decomposition. *Environ. Res.* **263**, 120285 (2024).
80. Machás, R., Santos, R. & Peterson, B. Elemental and stable isotope composition of *Zostera noltii* (Homeman) leaves during the early phases of decay in a temperate mesotidal lagoon. *Estuar. Coast. Shelf S.* **66**, 21–29 (2006).

81. Bradford, M. A., Tordoff, G. M., Eggers, T., Jones, T. H. & Newington, J. E. Microbiota, fauna, and mesh size interactions in litter decomposition. *Oikos* **99**, 317–323 (2002).
82. Blum, L. K. & Mills, A. L. Microbial-growth and activity during the initial stages of seagrass decomposition. *Mar. Ecol. Prog. Ser.* **70**, 73–82 (1991).
83. Peduzzi, P. & Herndl, G. J. Decomposition and significance of seagrass leaf litter (*Cymodocea-nodosa*) for the microbial food web in coastal waters (Gulf Trieste, Northern Adriatic Sea). *Mar. Ecol. Prog. Ser.* **71**, 163–174 (1991).
84. Newell, S. Y., Fell, J. W. & Miller, C. Deposition and decomposition of Turtlegrass leaves. *Int. Rev. Hydrobiol.* **71**, 363–369 (1986).
85. Chiu, S. H., Huang, Y. H. & Lin, H. J. Carbon budget of leaves of the tropical intertidal seagrass *Thalassia hemprichii*. *Estuar. Coast. Shelf Sci.* **125**, 27–35 (2013).
86. Ainsworth, E. A. & Gillespie, K. M. Estimation of total phenolic content and other oxidation substrates in plant tissues using Folin-Ciocalteu reagent. *Nat. Protoc.* **2**, 875–877 (2007).
87. Olson, J. S. Energy-storage and balance of producers and decomposers in ecological-systems. *Ecology* **44**, 322 (1963).
88. Trevathan-Tackett, S. M., Jeffries, T. C., Macreadie, P. I., Manojlovic, B. & Ralph, P. Long-term decomposition captures key steps in microbial breakdown of seagrass litter. *Sci. Total Environ.* **705**, 135806 (2020).
89. Aerts, R. Climate, leaf litter chemistry and leaf litter decomposition in terrestrial ecosystems: a triangular relationship. *Oikos* **79**, 439–449 (1997).
90. Golebiewski, M. et al. Rapid microbial community changes during initial stages of pine litter decomposition. *Microb. Ecol.* **77**, 56–75 (2019).
91. Wang, L. et al. Litter chemical traits strongly drove the carbon fractions loss during decomposition across an alpine treeline ecotone. *Sci. Total Environ.* **753**, 142287 (2021).
92. Ronkko, M., McIntosh, C. N., Antonakis, J. & Edwards, J. R. Partial least squares path modeling: time for some serious second thoughts. *J. Oper. Manag.* **47–48**, 9–27 (2016).
93. Chen, W. et al. Direct and indirect influences of long-term fertilization on microbial carbon and nitrogen cycles in an alpine grassland. *Soil Biol. Biochem.* **149**, 107922 (2020).
94. Yan, Y. et al. Climate-induced tree-mortality pulses are obscured by broad-scale and long-term greening. *Nat. Ecol. Evol.* **8**, 912–923 (2024).

Acknowledgements

This work was supported by the National Natural Science Foundation of China (42176155 and U1901221), the Guangdong Basic and Applied Basic Research Foundation (2025B1515020078), the Youth Innovation Promotion Association CAS (2023359), the Science and Technology Planning Project of Guangdong Province, China (2024B1212050003, 2023B1212060047), the National Key Research and Development Program of China (2022YFF0802201), the Hainan Province Science and Technology Special Fund (ZDYF2023SHFZ172), the Hainan Provincial Natural Science Foundation of China (425CXTD622), and. S.T-T. was supported by an ARC

DECRA Fellowship (DE210101029). We would also like to express our gratitude for the support from the Ocean Negative Carbon Emissions (ONCE) Program.

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Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s43247-025-02544-2>.

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Peer review information *Communications Earth and Environment* thanks and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editors: Haihan Zhang and Somaparna Ghosh [A peer review file is available].

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